```
100 'SKELETONIZATION PROGRAM
110 'Object pixels have a value of 0 (displayed as black)
120 'Background pixels have a value of 255 (displayed as white)
130 '
140 DIM X%[149,149]              'X%[  ,  ] holds the image being processed
150 '
160 GOSUB XXXX                   'Mythical subroutine to load X%[ , ]
170 '
180 FOR ITER% = 0 TO 5           'Run through six iteration loops
190 '
200   FOR R% = 1 TO 148          'Loop through each pixel in the image.
210   FOR C% = 1 TO 148          'Subroutine 5000 (Table 25-2) indicates which
220     GOSUB 5000               'pixels can be changed from black to white,
230   NEXT C%                    'by marking the pixels with a value of 1.
240   NEXT R%
250 '
260   FOR R% = 0 TO 149          'Loop through each pixel in the image changing
270   FOR C% = 0 TO 149          'the marked pixels from black to white.
280     IF X%(R%,C%) = 1 THEN X%(R%,C%) = 255
290   NEXT C%
300   NEXT R%
310 '
320 NEXT ITER%
330 '
340 END
```

TABLE 25-1

The decision to remove a pixel is based on four rules, as contained in the subroutine shown in Table 25-2. *All* of these rules must be satisfied for a pixel to be changed from black to white. The first three rules are rather simple, while the fourth is quite complicated. As shown in Fig. 25-12a, a pixel at location [R,C] has eight neighbors. The four neighbors in the horizontal and vertical directions (labeled 2,4,6,8) are frequently called the **close neighbors**. The diagonal pixels (labeled 1,3,5,7) are correspondingly called the **distant neighbors**. The four rules are as follows:

Rule one: The pixel under consideration must presently be black. If the pixel is already white, no action needs to be taken.

Rule two: At least one of the pixel's close neighbors must be white. This insures that the erosion of the thick ridges takes place from the outside. In other words, if a pixel is black, and it is completely surrounded by black pixels, it is to be left alone on this iteration. Why use only the *close neighbors*, rather than *all* of the neighbors? The answer is simple: running the algorithm both ways shows that it works better. Remember, this is very common in morphological image processing; trial and error is used to find if one technique performs better than another.

Rule three: The pixel must have more than one black neighbor. If it has only one, it must be the end of a line, and therefore shouldn't be removed.

Rule four: A pixel cannot be removed if it results in its neighbors being *disconnected*. This is so each ridge is changed into a continuous line, not a group of interrupted segments. As shown by the examples in Fig. 25-12,